Bank Capital Buffers and the Floor They Guard

Skeleton of 25 October 2026

1 Introduction

1.1 Question

Why does the standard dynamic model of bank capital put the buffers that US banks hold far below what they hold, and does a model that guards the floor regulation makes binding, with the risk that book capital carries, place them better?

1.2 Answer

The standard model, the dividend barrier of a bank whose capital moves with its own earnings and which fails at the legal minimum, predicts that banks pay dividends from thin buffers and almost never hold capital far above the reference, and observed banks do the opposite in every window scored (E1). The buffers banks hold match that model only when the risk they guard against is about three times their earnings volatility (E2). Measures built from losses, the supervisor's stress-test decline and the bank's own worst decline, put that risk at about one times earnings volatility, and the market's equity volatility at about eight times (E3).

A barrier model whose floor is the line the supervisor's restrictions start at, the well-capitalised ratio before 2010 and the minimum plus the conservation buffer after, and whose risk is the observed movement of book capital, beats the standard model on banks it had not been scored on (E4, E5), and on 2025, a year kept back from every choice (E6). It does not place the level of buffers on 2025, where it puts too few bank-quarters far above the reference (E7). The floor enters the barrier model only as a shift of the buffer (M3), so the floor changes where the barrier sits in the ratio and not the barrier itself (M2).

Two facts about buffers stay outside both models. Banks damp the movement of their own capital ratios by about a sixth through their capital actions (E8). Smaller banks hold larger buffers at the same risk, and neither access to a parent nor a listing accounts for it (E9).

Table 1: Headlines.

ID	Headline	Rows	Scope note
K1	The standard dividend-barrier model of bank capital fails both facts in every window, on US commercial banks 2005 to 2025.	E1, E6, M2	Scored on the two pre-registered facts only (the share of dividend quarters paid from a buffer below one earnings volatility over the reference, and the share of bank-quarters more than four above it). Peura and Keppo report a shortfall in level themselves (L1).
K2	Buffers imply a risk about three times earnings volatility, between the one times of loss-based measures and the eight times of the market.	E2, E3	Training window only, in the non-issuer barrier model. Risk scales, not mean capital shortfalls under a fixed scenario (L13). The implied volatility is Peura and Keppo's idea, at about 1.1 to 1.3 times in their sample (L15).
K3	A barrier model guarding the regulatory floor with book-capital risk beats the standard model out of sample and on a hold-out year.	E4, E5, E6, M3	Beats means a smaller sum of the two misses. The floor was set from sources after a floor near the reference had been seen to work (E5, E6). It does not place the level of buffers on the hold-out (E7).
K4	Banks' own capital actions damp the movement of their capital ratios, and smaller banks hold more capital at the same risk.	E8, E9	Description of observed data. The book-capital risk of K3 contains the damping, and removing it makes the model overshoot (E8).

2 Model

2.1 Players

Table 2: Players.

ID	Player	Choice faced	How the model settles it	In the literature	Rows
P1	Bank	When to pay dividends.	Derived. The bank pays at the dividend barrier, the point of smooth fit (M2).	Peura and Keppo's non-issuer bank, the case Milne and Whalley's buffer reduces to (L1, L3).	M2, L1, L3
P2	Supervisor	Where to set the floor and when to restrict a bank.	Taken as given. The floor is a rule (A5), with its post-2009 value from the conservation buffer (M5).	Orozco and Rubio's well-capitalised threshold and its replacement after 2009 (L7, L8).	M5, L7, L8
P3	Insurer	How to price deposit insurance.	Not modelled. The insurer is state, entering only through the floor banks guard.	The lower assessment above the well-capitalised line is one reason banks guard it (L7).	L7
P4	Investors	How to value bank equity.	Not modelled. Investors enter only as one measure of the risk the bank guards against (E3).	Craine and Martin find market capital moves far more than book capital (L12).	E3, L12

2.2 Assumptions

Table 3: Assumptions.

ID	Assumption	In plain English	Symbols	Model rows	Reason
A1	$dx_t = \mu_i\,dt + s_i\,dW_t - dD_t$.	The buffer over the floor drifts up with the bank's mean earnings, moves with the judged risk through a Brownian motion, and falls by the dividends paid.	x, μ_i, s_i, W_t, D_t	M1, M2, M6	Peura and Keppo's capital process (Appendix D, process).
A2	Dividends are paid only at the barrier u_0, and no equity is issued.	The bank pays out at the dividend barrier and never raises capital.	u_0	M2, M6	The non-issuer case, an upper bound on the buffer when issuance is costly (L3; Appendix D, issuance).
A3	The bank is closed when $x_t = 0$.	The bank fails when the buffer over the floor is used up.	x	M1, M2, M3, M6	Closure at the floor (Appendix D, closure).
A4	The bank maximises its expected dividends D_t discounted at $\rho = 0.04$ up to τ.	The bank maximises its expected dividends, discounted at the discount rate of four percent a year, its cost of equity, up to the closure time.	ρ, D_t, τ	M1, M6	Peura and Keppo's value (Appendix D, discounting).
A5	$\underline{k} = 0.08$ in the baseline, and $\underline{k} = 0.10$ to 2009Q4 and $\underline{k} = 0.105$ from 2010Q1 in the enhanced model.	The regulatory floor is the legal minimum in the baseline, and the well-capitalised line and then the minimum plus the conservation buffer in the enhanced model.	$\underline{k}$	M3, M5	The lines the sources find binding (L7, L8, L9; Appendix D, floor).
A6	$s_i = \sigma_i$ in the baseline and $s_i = \sigma_{D,i}$ in the enhanced model.	The judged risk is the earnings volatility in the baseline and the book-capital volatility in the enhanced model.	s_i, σ_i, $\sigma_{D,i}$	M1, M2, M4	Appendix D, judged risk.

2.3 Results

Table 4: Model results.

ID	Claim	Statement	Lean	Proof
M1	The value equation has one positive and one negative root.	$s \neq 0$, $\rho > 0 \Rightarrow$ $p = d_+(\mu, s, \rho) > 0 > m = d_-(\mu, s, \rho)$, both roots of $\frac{1}{2}s^2d^2 + \mu d - \rho = 0$.	`PeuraKeppo.char_eq`, `PeuraKeppo.dplus_pos`, `PeuraKeppo.dminus_neg`	p. 8
M2	The dividend barrier satisfies smooth fit and equals Milne and Whalley's buffer.	$p > 0 > m \Rightarrow a_1 e^{-pu_0} + a_2 e^{-mu_0} = 0$ at $u_0 = \frac{2}{p-m} \log \frac{-m}{p}$, with $a_1 > 0 > a_2$, and $\frac{1}{p-m} \log \frac{m^2}{p^2} = u_0$.	`PeuraKeppo.no_issue_barrier`, `PeuraKeppo.coeff_signs`, `MilneWhalley.buffer_eq_u0`	p. 8
M3	Raising the floor shifts the buffer and leaves its movements unchanged.	$(C_t + d) - (C_0 + d) = C_t - C_0$ for every d.	`MilneWhalley.buffer_shift_invariant`	p. 9
M4	A stress-test decline over nine quarters converts to a judged risk.	$\Delta/(z\sqrt{9/4}) = \Delta/(\frac{3}{2}z)$, and $k^{\min} \leq k^{\text{act}} \Rightarrow \Delta = k^{\text{act}} - k^{\min} \geq 0$.	`StressTests.judged_sigma`, `StressTests.sqrt_horizon`, `StressTests.decline_of_rows`	p. 9
M5	After 2009 the floor is the reference and lies above the well-capitalised line.	$0.08 + 0.025 = 0.105$ and $0.10 < 0.105$.	`OrozcoRubio.basel3_floor_eq_R`, `OrozcoRubio.basel3_floor_above_pca`	p. 9
M6	The value function of the non-issuer bank solves the master equation, a variational inequality in dividends, with Peura and Keppo's closed form and the barrier u_0.	$\max\{\frac{1}{2}s^2V'' + \mu V' - \rho V, 1 - V'\} = 0$, $V(0) = 0$. For $x < u_0$, $V(x) = a_1 e^{-p(u_0 - x)} + a_2 e^{-m(u_0 - x)}$ solves $\frac{1}{2}s^2V'' + \mu V' - \rho V = 0$, with $V(0) = 0$, $V'(u_0) = 1$, $V''(u_0) = 0$ and $V(u_0) = \mu/\rho$. For $x \geq u_0$, $V(x) = \mu/\rho + x - u_0$ and $\frac{1}{2}s^2V'' + \mu V' - \rho V \leq 0$.	`EnhancementMap.continuation_solves`, `PeuraKeppo.no_issue_barrier`, `PeuraKeppo.smooth_paste`, `PeuraKeppo.value_at_barrier`, `EnhancementMap.payout_region`	p. 9

3 Empirical analysis

Every row below is a results file that the checker re-runs on its pinned data, committed after the pre-registrations it names. F1 is the share of dividend quarters paid from a buffer below σ_i over R, and F2 the share of bank-quarters more than $4\sigma_i$ above R. The miss of a model is the sum of its two distances to the observed shares, in percentage points.

Table 5: Empirical rows.

ID	Claim	Verdict	Model rows	Map rows	Evidence
E1	The baseline misses both facts in both windows. On 2015 to 2024 for the 61 largest banks it predicts F1 90.0% against 4.8% and F2 0.0% against 61.6%.	holds	M2	thin-buffer dividend share, far-above share	`comparison/PREREG_comparison.md`, `comparison/PREREG_comparison_2.md`, `comparison/results_pk.md`
E2	In the non-issuer barrier model the training-window F2 of the 61 largest banks (48.0%) is reached when the judged risk is about three times earnings volatility (42.8% at three times, minimum 8%).	holds	M2	earnings volatility, judged risk, far-above share	`comparison/PREREG_comparison_6.md`, `comparison/PREREG_comparison_2.md`, `comparison/results_scale.md`

ID	Claim	Verdict	Model rows	Map rows	Evidence
E3	Stress-test declines give a judged risk with median 0.99 times earnings volatility and the bank's own worst decline 1.00 times, equity volatility scaled by the capital ratio 7.75 times.	holds	M4	judged risk, market judged risk, stress-test judged risk, own-loss judged risk	comparison/PREREG_comparison_7.md, comparison/PREREG_comparison_8.md, comparison/PREREG_comparison_9.md, comparison/PREREG_comparison_2.md, comparison/results_judged.md, comparison/PREREG_comparison_16.md, comparison/results_own_loss.md
E4	With equity volatility as the judged risk the barrier model beats the baseline on 2015 to 2024 on the 17 stress-tested banks (miss 30.0 against 139.8) and on 18 further banks scored blind (32.8 against 149.3).	holds	M2	market judged risk, thin-buffer dividend share, far-above share	comparison/PREREG_comparison_10.md, comparison/PREREG_comparison_7.md, comparison/PREREG_comparison_8.md, comparison/PREREG_comparison_9.md, comparison/results_test_judged.md, comparison/PREREG_comparison_12.md, comparison/PREREG_comparison_13.md, comparison/PREREG_comparison_14.md, comparison/results_extend.md
E5	With book-capital volatility and a 10% floor the barrier model beats the baseline on 26 banks scored blind on 2015 to 2024 (miss 15.2 against 149.2).	holds	M2, M3	regulatory floor, book-capital volatility, thin-buffer dividend share, far-above share	comparison/PREREG_comparison_17.md, comparison/results_book_vol.md
E6	The frozen model, with book-capital volatility and the floor of A5, beats the baseline on the 2025 hold-out for 42 banks (miss 23.3 against 182.4) and agrees on F1 (0.2% against 1.6%).	holds	M2, M3, M5	regulatory floor, book-capital volatility, thin-buffer dividend share, far-above share	comparison/PREREG_comparison_20.md, comparison/results_holdout.md, comparison/results_frozen.md
E7	The frozen model matches the level of buffers on the 2025 hold-out. It predicts F2 67.1% against 89.1% observed, 22.0 points low.	fails	M2, M3, M5	far-above share	comparison/PREREG_comparison_20.md, comparison/results_holdout.md
E8	The capital ratio before the bank's own capital actions moves 1.19 times as much as the ratio after them (median, the 61 largest banks), and with that risk the model overshoots F2 (71.2% against 48.0%).	holds	M2	book-capital volatility, do-nothing volatility	comparison/PREREG_comparison_19.md, comparison/results_do_nothing.md
E9	Across banks of all sizes the median buffer falls with bank size (Spearman -0.34 and -0.39) and rises with risk ($+0.23$, $+0.26$). Neither model reproduces the size gradient, and stand-alone banks and banks without a listed parent hold no more than others of the same size.	holds	M2	buffer over the floor, earnings volatility, bank size	comparison/PREREG_comparison_17.md, comparison/results_book_vol.md, comparison/PREREG_comparison_18.md, comparison/results_access.md

4 Literature

Table 6: Literature read.

ID	Paper	What we rely on	Quote	Page	Rows
L1	Peura and Keppo [2006]	The baseline's own authors report it under-explains the level of capital (E1). Their fit used their own barriers, not common facts.	"The model falls somewhat short of explaining the level of capital held by banks."	2188	M2, C1
L2	Peura and Keppo [2006]	Their best fit, on a subsample (E1 scores all banks).	"the model explains some 40% of the variation in capital ratios across banks and 70% of the average level of the capital buffer."	2191	C1
L3	Milne and Whalley [2001]	The barrier without issuance is an upper bound on the buffer when issuance is costly (A2, M2).	"so the closed form solution represents an upper bound on the capital buffer."	42	M2
L4	Berger et al. [2008]	Targets far above the regulatory line (E1, E2).	"choose target capital levels substantially above well-capitalized regulatory minima"	0	C1
L5	Berger et al. [2008]	Targets rise with risk and fall with size (E9).	"these targets increase with BHC risk but decrease with BHC size"	0	C7
L6	Berger et al. [2008]	Their explanations of the size gradient, offered and not separated (E9 tests one).	"Finally, bank capital ratios tend to be inversely related to asset size. This might reflect greater diversification, scale economies in risk management, and/or lower expected cost of raising new equity on short notice."	7	C7
L7	Orozco and Rubio [2025]	Banks guard the well-capitalised line before 2010 (A5). Their evidence, not ours.	"a significant discontinuity at the 10% regulatory capital threshold, where banks receive benefits for exceeding it."	1421	M5, C3, C4
L8	Orozco and Rubio [2025]	The post-2009 floor (A5, M5).	"That is, this new buffer increases the minimum reg-ulatory capital to at least 10.5%."	1430	M5
L9	Orozco and Rubio [2025]	The line that binds is 10%, not 8%, their reading of Gropp and Heider (L11).	"Our results align with the notion that capital is costly but suggest that the 10% ratio is the relevant minimum, elucidat-ing why previous studies find weak evidence of binding capital requirements at the 8% regulatory minimum"	1425	M5
L10	Orozco and Rubio [2025]	Their design is local to the line, so the level of buffers far above it (F2) is not theirs.	"Our main sample is restricted to banks that reported regulatory capital between 8% and 12%."	1427	C3
L11	Gropp and Heider [2009]	Regulation second order for large listed banks, tested at the Basel Tier 1 minimum (L9 answers).	"The paper shows that mispriced deposit insurance and capital regulation were of second order importance in determining the capital structure of large U.S. and European banks during 1991 to 2004."	4	C4
L12	Craine and Martin [2017]	Book capital moves little, market capital much (E3). Theirs is dispersion across banks at one date (E8 is over time).	"Book value capital ratios hardly move in response to severe economic distress."	1	C2
L13	Craine and Martin [2017]	The stress test and the market call for similar average changes in capital (K2 is a risk scale).	"is close to the average deviation for the market value measures (columns 5 and 6)."	14	C2
L14	Lang and Menno [2026]	A buffer requirement above the minimum binds in theory (A5). A two-period model with no buffers far above it.	"Banks do not maintain voluntary capital buffers above the CBR"	18	M5
L15	Peura and Keppo [2006]	They introduce the implied volatility that reproduces buffers, about 1.3 times their estimate for large banks on 1983 to 2002 (C6 measures it again, larger).	"For large banks (average assets over 40 billion US$), the average capital buffer of 3.8% implies a volatility of 1.06%."	2190	C6
L16	Peura and Keppo [2006]	The size gradient beyond volatility, which they assign to capital-market access (C7 tests that reading).	"suggesting that larger banks hold less capital than small banks, even after controlling for possible differences in their"	2190	C7

ID	Paper	What we rely on	Quote	Page	Rows
L17	Dai et al. [2019]	Two versions of one dynamic model compared out of sample on US banks, scored on equity value, dividends and recapitalisations (C5 compares models of different authors on buffer facts).	"the partially observed model explains the out-of-sample variations substantially better than the fully observed model."	86	C5
L18	Dai et al. [2019]	Reported volatility understates the volatility behind it, the nearest antecedent of C6 and of the damping in E8.	"they are able to hide about one-third of the true underlying asset volatility."	84	C6
L19	Juselius et al. [2025]	A buffer of about 3.8 standard deviations of buffer shocks after the crisis, for 34 large groups. The denominator is the volatility of buffer changes net of the bank's actions, not earnings volatility, and no model is asked to reproduce the buffer (C6).	"The implication is that an eight-standard-deviation shock is needed to wipe out the MB target on impact in the former regime, whereas a four-standard-deviation shock would have a similar impact in the latter regime."	12	C6
L20	Corbae and D'Erasmo [2021]	A dynamic model of US banks estimated to buffer levels by size in sample, 1984 to 2007 (C5 predicts without fitting, out of sample).	"After estimating parameters to match observed levels of buffers in both the imperfect and perfect competition economies"	31	C5
L21	Corbae and D'Erasmo [2021]	Funding volatility as the size channel, calibrated and not tested (C7 has not tested it).	"big banks have less volatile funding inflows, which is one important factor explaining why they hold a smaller capital buffer in our model."	24	C7
L22	Sengupta and Hogue [2014]	Banks with no holding company hold more Tier 1 capital within banks, community banks 1996 to 2006. E9 pools one-bank holding companies with them and compares across banks of a size.	"Non-BHC banks are associated with a 1.06-percentage-point higher Tier 1 leverage ratio and a 2.19-percentage-point higher Tier 1 risk-based capital ratio, after controlling for other determinants of capital ratios."	16	C7
L23	Peura and Jokivuolle [2003]	They invert a buffer model on the median US capital ratio for a confidence level, with the risk model fixed (C6 inverts for a risk scale with the decision rule fixed).	"Given a bank's portfolio and its capital ratio, there is an implied value of a which makes the capital ratio solved from the model equal to the observed capital ratio."	20	C6
L24	Peura and Jokivuolle [2003]	A deterministic stress scenario carries no probability, so the quantile z of M4 is assumed.	"The obvious shortcoming of such an analysis is the absence of any likelihoods attached to the chosen rating transition scenario itself."	36	M4
L25	Peura and Keppo [2006]	Their Proposition 2, the variational inequalities of M6 for their bank with issuance, by standard arguments.	"Now the following characterization of optimum can be established using standard arguments"	2170	M6
L26	Peura and Keppo [2006]	Without issuance their value function and policy are those of M6, and their proof shows the function is the value function (A2).	"the optimal policy and the value function reduce to those of the Milne and Robertson (1996) model without the capital issue option."	2175	M6

Papers cited and not read are not used.

5 Conclusions

Table 7 holds the conclusions that rest on model rows, empirical rows and papers read in full. Table 8 holds every claim that something is new and still waits on something, and what.

Table 7: Conclusions.

ID	Statement	Rows
C1	The standard dividend-barrier model of bank capital fails on US commercial banks because it guards the legal minimum with the bank's earnings volatility, and observed buffers sit where a risk three times larger would put them.	E1, E2, L1
C2	Neither loss-based measure, the stress test or the bank's own worst decline, supplies that risk, and the market's equity volatility overshoots it.	E3, M4
C3	Guarding the floor that regulation makes binding, with the movement of book capital as the risk, beats the standard model on banks and on a year it had not seen, without placing the level of buffers on that year.	E4, E5, E6, E7, M3, L7, L8
C4	The floor the enhanced model guards rests on sources rather than on fit, the well-capitalised line before 2010 and the minimum plus the conservation buffer after.	M5, L7, L8, L9, L14
C6	Observed buffers imply a risk about three times earnings volatility on 2005 to 2014, against Peura and Keppo's 1.1 to 1.3 times on their 1983 to 2002 subsample (L15), with loss-based measures at about one and the market at about eight. Juselius, Ponte Marques and Tarashev's buffer of about four standard deviations of buffer shocks has a different denominator (L19), and Peura and Jokivuolle invert buffers for a confidence level rather than a risk scale (L23). No paper read measures this risk scale against loss-based and market measures.	E2, E3, L13, L15, L18, L19, L23

Table 8: Pending.

ID	Statement	Rows
C5	Earlier work fits a single model to the data it then explains, Peura and Keppo by calibration and Corbae and D'Erasmo by estimation to buffer levels in sample (L1, L20), and Dai, Huang and Keppo compare two versions of one model out of sample on other targets (L17). No paper read scores the published dynamic models of different authors against each other on pre-registered buffer facts out of sample, with no parameter fitted to those facts. *Pending:* Migueis (2017), a recalibration of Peura and Keppo, seen in abstract only.	L1, L17, L20
C7	The size gradient of buffers survives controls for access to capital by affiliation and listing, the explanation Peura and Keppo give for it (L16). Sengupta and Hogue find banks without a holding company hold more within banks on another definition and period (L22), and Corbae and D'Erasmo assign the gradient to funding volatility and issuance costs without testing either (L21). *Pending:* a test of funding volatility, and a split of stand-alone banks into those with no holding company and those of one-bank holding companies, each under its own pre-registration.	E9, L6, L16, L21, L22

References

Allen N. Berger, Robert DeYoung, Mark J. Flannery, David Lee, and Özde Öztekin. How do large banking organizations manage their capital ratios? *Journal of Financial Services Research*, 34(2–3):123–149, 2008.

Dean Corbae and Pablo D'Erasmo. Capital buffers in a quantitative model of banking industry dynamics. Working Paper 21-24, Federal Reserve Bank of Philadelphia, June 2021.

Roger Craine and Vance L. Martin. Safe capital ratios for bank holding companies. Revised draft, July 2017, 2017.

Min Dai, Shan Huang, and Jussi Keppo. Opaque bank assets and optimal equity capital. *Journal of Economic Dynamics and Control*, 100:369–394, 2019. Read as chapter 2 of S. Huang, Two Stochastic Control Problems in Capital Structure and Portfolio Choice, NUS PhD thesis, arXiv:2107.02242v1.

Reint Gropp and Florian Heider. The determinants of bank capital structure. Working Paper 1096, European Central Bank, September 2009.

Mikael Juselius, Aurea Ponte Marques, and Nikola Tarashev. Banks' regulatory risk tolerance. Version of 20 October 2025; also BIS Working Paper 1287, 2025.

Jan Hannes Lang and Dominik Menno. A structural model of capital buffer usability. Working Paper 3188, European Central Bank, 2026.

Alistair Milne and A. Elizabeth Whalley. Bank capital regulation and incentives for risk-taking. Technical report, City University Business School, December 2001. Revised version of Bank of England Working Paper 90 (1999).

Luciana Orozco and Silvina Rubio. Regulatory capital management to exceed thresholds. *Journal of Money, Credit and Banking*, 57(6):1421–1464, 2025. doi: 10.1111/jmcb.13230.

Samu Peura and Esa Jokivuolle. Simulation-based stress testing of banks' regulatory capital adequacy. Discussion Papers 4/2003, Bank of Finland, 2003.

Samu Peura and Jussi Keppo. Optimal bank capital with costly recapitalization. *Journal of Business*, 79(4): 2163–2201, 2006.

Rajdeep Sengupta and Eric W. Hogue. The effect of risk and organizational structures on bank capital ratios. *Federal Reserve Bank of Kansas City Economic Review*, (Fourth Quarter), 2014.

A Proofs

Notation. x is the buffer over the floor, $\underline{k}$ the floor, $R = 0.105$ the reference, μ_i and σ_i the mean and the volatility of bank i's yearly earnings, s_i the judged risk, ρ the discount rate, $p = d_+(\mu, s, \rho)$ and $m = d_-(\mu, s, \rho)$ the roots, $a_1 = m/(pm - p^2)$, $a_2 = p/(pm - m^2)$, u_0 the dividend barrier, V the value function, D_t cumulative dividends, τ the closure time, Δ a stress-test decline in the capital ratio and z a normal quantile.

Proof of M1

Lean: `PeuraKeppo.char_eq`, `PeuraKeppo.dplus_pos`, `PeuraKeppo.dminus_neg`.

$$d_\pm = \frac{-\mu \pm \sqrt{\mu^2 + 2\rho s^2}}{s^2}, \qquad \tfrac{1}{2} s^2 d_\pm^2 + \mu d_\pm - \rho = 0.$$

With $\rho > 0$, $\sqrt{\mu^2 + 2\rho s^2} > |\mu|$, so $d_+ > 0$ and $d_- < 0$.

□

Proof of M2

Lean: `PeuraKeppo.no_issue_barrier`, `PeuraKeppo.coeff_signs`, `MilneWhalley.buffer_eq_u0`.

$$e^{(p-m)u_0} = \left(\frac{-m}{p}\right)^2, \quad a_1 e^{-pu_0} + a_2 e^{-mu_0} = e^{-pu_0}\left(a_1 + a_2 \frac{m^2}{p^2}\right) = 0,$$

$$a_1 = \frac{m}{p(m-p)} > 0 > a_2 = \frac{p}{m(p-m)}, \qquad \frac{1}{p-m} \log \frac{m^2}{p^2} = \frac{2}{p-m} \log \frac{-m}{p}.$$

□

Proof of M3

Lean: `MilneWhalley.buffer_shift_invariant`.

$$(C_t + d) - (C_0 + d) = C_t - C_0,$$

so with $k_t = \underline{k} + x_t$ a change $\underline{k} \mapsto \underline{k} + d$ moves k_t by d and leaves $x_t - x_0$, u_0 and the closure time $\inf\{t : x_t = 0\}$ unchanged.

□

Proof of M4

Lean: `StressTests.judged_sigma`, `StressTests.sqrt_horizon`, `StressTests.decline_of_rows`.

$$\sqrt{9/4} = \tfrac{3}{2}, \qquad \frac{\Delta}{z\sqrt{9/4}} = \frac{\Delta}{\frac{3}{2}z}, \qquad k^{\min} \le k^{\text{act}} \;\Rightarrow\; \Delta = k^{\text{act}} - k^{\min} \ge 0.$$

□

Proof of M5

Lean: `OrozcoRubio.basel3_floor_eq_R`, `OrozcoRubio.basel3_floor_above_pca`.

$$0.08 + 0.025 = 0.105 = R, \qquad 0.10 < 0.08 + 0.025.$$

□

Proof of M6

Lean: `EnhancementMap.continuation_solves`, `PeuraKeppo.no_issue_barrier`, `PeuraKeppo.smooth_paste`, `PeuraKeppo.value_at_barrier`, `EnhancementMap.payout_region`.

With $b_1 = a_1 e^{-pu_0}$, $b_2 = a_2 e^{-mu_0}$ and $V = b_1 e^{px} + b_2 e^{mx}$,

$$V' = b_1 p e^{px} + b_2 m e^{mx}, \quad V'' = b_1 p^2 e^{px} + b_2 m^2 e^{mx}, \quad \tfrac{1}{2}s^2 V'' + \mu V' - \rho V = \sum_j b_j e^{d_j x}\left(\tfrac{1}{2}s^2 d_j^2 + \mu d_j - \rho\right) = 0.$$

$$V(0) = a_1 e^{-pu_0} + a_2 e^{-mu_0} = 0, \qquad V'(u_0) = a_1 p + a_2 m = 1,$$
$$V''(u_0) = a_1 p^2 + a_2 m^2 = 0, \qquad V(u_0) = a_1 + a_2 = \mu/\rho.$$
$$x \ge u_0 \;\Rightarrow\; \tfrac{1}{2}s^2 \cdot 0 + \mu \cdot 1 - \rho\left(\frac{\mu}{\rho} + x - u_0\right) = -\rho(x - u_0) \le 0.$$

□

B Evidence and verification

The Lean theorems are those of `lean/mathlib/`, built and audited by the parent bundle (`verify.sh`). The checker `enhancement/checks/verify_enhancement.py` applies MS-1 to MS-9 and re-runs every results file an empirical row cites.

What Lean does not carry. The Brownian motion of A1, that the function of M6 is the value function and that paying at u_0 is optimal, which is Peura and Keppo's result (L25, L26), and the simulation of the facts. The facts are computed on pinned data and reproduced exactly by the checker, which is the third class of evidence of this bundle.

Illustrations. No row rests on SymPy or sampling outside a pinned, reproduced results file.

C Terminology and measurement map

The map names every object of the model once, and the manuscript uses these names and no others. Inputs are what the model takes as given, internal objects are built from the inputs inside the model, and outputs are what the rows predict. Directly observed is Yes when a record of the world gives the object without computation, which is when the last column begins with Recorded. Every computed object names its recovery formula and the Lean theorem that carries it. The same tables make the standalone `enhancement/MEASUREMENT_MAP.tex`.

C.1 Inputs

Table 9: Inputs of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
k	**capital ratio**	A real number above zero.	Total capital, tier 1 plus tier 2, over risk-weighted assets, for each bank and quarter.	No	Computed as `(RBCT1J+RBCT2)/RWAJT` from the Call Report, fields defined in the record lit/fdic_2026. The ratio is free of the unit of account (`EnhancementMap.capital_ratio_units`).
$\underline{k}$	**regulatory floor**	A real number between zero and one.	The capital ratio at which the bank is closed in the model (A3, A5).	No	Assumed by A5 at the lines the sources find binding, 8% for the baseline, and 10% to 2009Q4 and 10.5% from 2010Q1 for the enhanced model (M5).
R	**reference**	A real number, 0.105.	The ratio the facts are measured from.	No	Assumed, fixed in comparison/PREREG_comparison.md before any fact was computed.
μ_i	**mean earnings**	A real number.	The mean of the bank's yearly net income over the previous year's risk-weighted assets, less the riskless return on its capital ratio (A1).	No	Computed over ten years of December reports by the code of replication/peura_keppo_2006, each year as net income over risk-weighted assets less the riskless rate times the capital ratio (`EnhancementMap.net_earnings`).
σ_i	**earnings volatility**	A positive real number.	The standard deviation of the same yearly earnings (A6).	No	Computed over the same ten years from the same yearly terms (`EnhancementMap.net_earnings`).
ρ	**discount rate**	A positive real number, 0.04.	The rate at which the bank discounts dividends, its cost of equity (A4).	No	Assumed at Peura and Keppo's value.
s_i	**judged risk**	A positive real number.	The volatility the buffer over the floor moves with in the model (A1, A6).	No	Assumed by A6 to be the earnings volatility or the book-capital volatility, or one of the measures of E3.
$\sigma_{D,i}$	**book-capital volatility**	A positive real number.	The standard deviation of the yearly change in the capital ratio over ten years (A6).	No	Computed by comparison/book_vol.py. The change is the same for the capital ratio and for the buffer over the floor (`EnhancementMap.change_free_of_floor`).
$\sigma_{E,i}$	**equity volatility**	A positive real number.	The annualised standard deviation of daily log changes of the parent's share price.	No	Computed from pinned daily prices by comparison/judged_risk.py as $\sqrt{252}$ times the daily standard deviation (`EnhancementMap.annualise`).

Symbol	Name	Type	Represents	Directly observed	Obtained how
X_i	**mean capital ratio**	A real number above zero.	The mean of the bank's capital ratio over ten December reports.	No	Computed by comparison/judged_risk.py; a mean lies between the least and the largest ratio (`EnhancementMap.mean_between`).
$\sigma_{B,i}$	**market judged risk**	A positive real number.	The volatility of the capital ratio if book capital moved with the market value of equity (E3).	No	Computed as $\sigma_{E,i}X_i$, the standard deviation of X_i times a return (`EnhancementMap.market_scale`).
Δ	**stress-test decline**	A real number at least zero.	The fall in a holding company's tier 1 common ratio from its actual value to its projected minimum under the severely adverse scenario.	No	Computed from the published rows of lit/dfast_2013 and lit/dfast_2014 as the actual less the minimum (`StressTests.decline_of_rows`).
z	**normal quantile**	A positive real number.	The quantile of a normal loss that a decline is read as (M4).	No	Assumed on the grid 1.645, 2.326, 3.090, with 2.326 central.
$\sigma_{A,i}$	**stress-test judged risk**	A positive real number.	The volatility under which a nine-quarter decline Δ is the z-quantile (E3).	No	Computed as $\Delta/(\frac{3}{2}z)$ (`StressTests.judged_sigma`).
$\sigma_{C,i}$	**own-loss judged risk**	A positive real number.	The same conversion applied to the bank's own worst nine-quarter decline (E3).	No	Computed by comparison/own_loss.py as $\Delta/(\frac{3}{2}z)$ with the bank's decline (`StressTests.judged_sigma`).
$\sigma_{N,i}$	**do-nothing volatility**	A positive real number.	The standard deviation of the yearly change in Berger et al.'s do-nothing ratio (E8).	No	Computed by comparison/do_nothing.py; each change is an earnings part plus an asset part (`EnhancementMap.do_nothing_split`).
S_i	**bank size**	A real number.	The log of the bank's mean total assets (E9).	No	Computed from the Call Report field `ASSET`; the log preserves the order of sizes (`EnhancementMap.size_monotone`).

C.2 Internal objects

Table 10: Internal objects of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
x	**buffer over the floor**	A real number, the state.	The capital ratio less the regulatory floor (A1, A3).	No	Computed as $k - \underline{k}$ (`EnhancementMap.buffer_of_ratio`).
W_t	**Brownian motion**	A standard Brownian motion.	The source of the buffer's random movement (A1).	No	Never observed.
p	**positive root**	A positive real number.	The positive root of the value equation (M1).	No	Computed from μ_i, s_i and ρ (`PeuraKeppo.dplus_pos`).
m	**negative root**	A negative real number.	The negative root of the value equation (M1).	No	Computed from μ_i, s_i and ρ (`PeuraKeppo.dminus_neg`).

Symbol	Name	Type	Represents	Directly observed	Obtained how
a_1, a_2	**barrier coefficients**	Two real numbers of opposite sign.	The weights of the two exponentials in the smooth-fit condition (M2).	No	A device of the proof of M2 (`PeuraKeppo.coeff_signs`).
u_0	**dividend barrier**	A positive real number.	The buffer over the floor at which dividends are paid (A2, M2).	No	Computed as $\frac{2}{p-m}\log\frac{-m}{p}$ (`PeuraKeppo.no_issue_barrier`).
V	**value function**	A function of the buffer over the floor.	The value of the bank's discounted dividends under the best dividend policy (A4, M6).	No	Computed in closed form by M6, below the barrier as a solution of the continuation equation (`EnhancementMap.continuation_solves`).
τ	**closure time**	A stopping time.	The first time the buffer over the floor reaches zero (A3, A4).	No	Never observed for the banks scored, which survive the windows.

C.3 Outputs

Table 11: Outputs of the model.

Symbol	Name	Type	Represents	Directly observed	Obtained how
F_1	**thin-buffer dividend share**	A share between zero and one.	The share of quarters with dividends paid from a buffer over the reference below σ_i, observed or predicted.	No	Computed by comparison/facts.py and window.py on the data, and by the simulations on the model (`EnhancementMap.share_unit`).
F_2	**far-above share**	A share between zero and one.	The share of bank-quarters more than $4\sigma_i$ above the reference, observed or predicted.	No	Computed by the same scripts (`EnhancementMap.share_unit`).
D_t	**dividends**	A nondecreasing process.	Cumulative dividends, the bank's control (A1, A4, M6).	Yes	Recorded in the Call Report as dividends declared, `EQCDIV`.

D Primitives and scope

Process. A1 is Peura and Keppo's process for capital over the minimum, with the bank's own mean and volatility. M1 and M2 depend on it.

Issuance. A2 drops issuance. Milne and Whalley show the barrier without issuance bounds the buffer with costly issuance from above (L3), so the enhanced model's buffer is the largest the family gives. M2 depends on it.

Closure. A3 closes the bank at the floor. The facts are measured from R, not from the floor, so the floor moves the barrier in the ratio (M3) without changing the facts' yardstick.

Discounting. A4 takes Peura and Keppo's cost of equity of four percent. The training-window scale check (E2) reports lower values.

Floor. A5 is the author's choice of 25 October 2026, taken from Orozco and Rubio (L7, L8, L9) and Lang and Menno (L14). It was made after a floor near R had been seen to work on 2015 to 2024, which is why the 2025 hold-out (E6, E7) decides the frozen model.

Judged risk. A6 replaces the earnings volatility by the book-capital volatility, which includes the bank's own capital actions (E8). The alternatives scored and rejected, the stress-test decline, the own worst decline and the equity volatility, are E3. Converting a decline to a volatility assumes a normal quantile z, and a stress scenario has no probability of its own (L24). Credit losses are long-tailed, so a normal quantile if anything overstates the volatility a decline implies, and the stress-test judged risk of E3 is an upper value.